Life Sciences - Structural Correspondence with the 5-Stage Model

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Life Sciences

Survey of structural correspondence with the 5-stage model (Field · Wave · Edge · Vortex · Bundle)

Survey Overview

- **Scope**: 11 major theories in life sciences
- **Research question**: Do life-science theories correspond structurally to the 5-stage model?
- **Results**: Strong correspondence in 1 case, conditional correspondence in 1 case

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Overview of the 5-Stage Model

Stage	Definition
Field (ba)	Undifferentiated state. Initial condition in which neither direction nor structure has yet been determined
Wave (nami)	Exploration stage in which multiple directions diverge and compete
Edge (en)	Tension state in which opposing elements coexist without converging on either side. The place where things meet at a boundary, influence each other, and relationships are formed
Vortex (uzu)	Stage in which new coherence (order) spontaneously emerges from tension
Bundle (taba)	Stage in which form is fixed and stabilized as a reusable structure

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Overview of Structural Correspondences

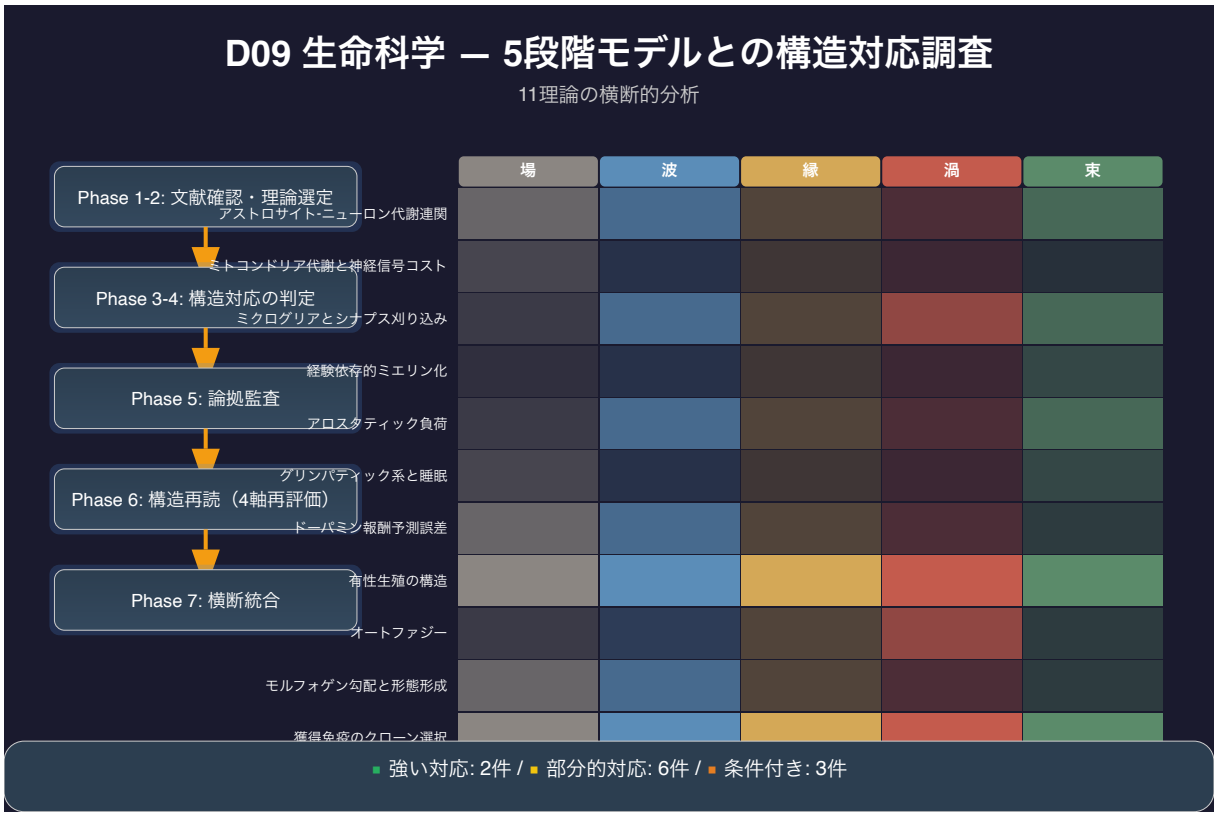

Figure 1: Domain — Research Overview Infographic

Temperature	Theory / Concept	Positioning
Near-confirmed	Structure of sexual reproduction, clonal selection of adaptive immunity, morphogen gradients and morphogenesis	Correspondence across all 5 stages is specific and the sequential ordering of the process is clear
Well-supported	Microglial synaptic pruning, autophagy, dopamine reward prediction error, allostatic load	Correspondence exists but is skewed toward specific stages or maps onto processes running in the opposite direction (decomposition / failure) from “typical creation”
Conditional	Astrocyte-neuron metabolic coupling, experience-dependent myelination, glymphatic system	Correspondence is centered on specific stages (stabilization toward Bundle, reset toward Field) rather than the process as a whole
Cost structure	Mitochondrial metabolism and neural signal cost	Rather than corresponding to any particular stage, this describes the physical constraints within which all stages can occur

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Key Entry 1: Structure of Sexual Reproduction

- Sexual reproduction is an innovation that arose early in eukaryotic evolution and remains the mode of reproduction maintained by the majority of multicellular organisms on Earth today. Genetic

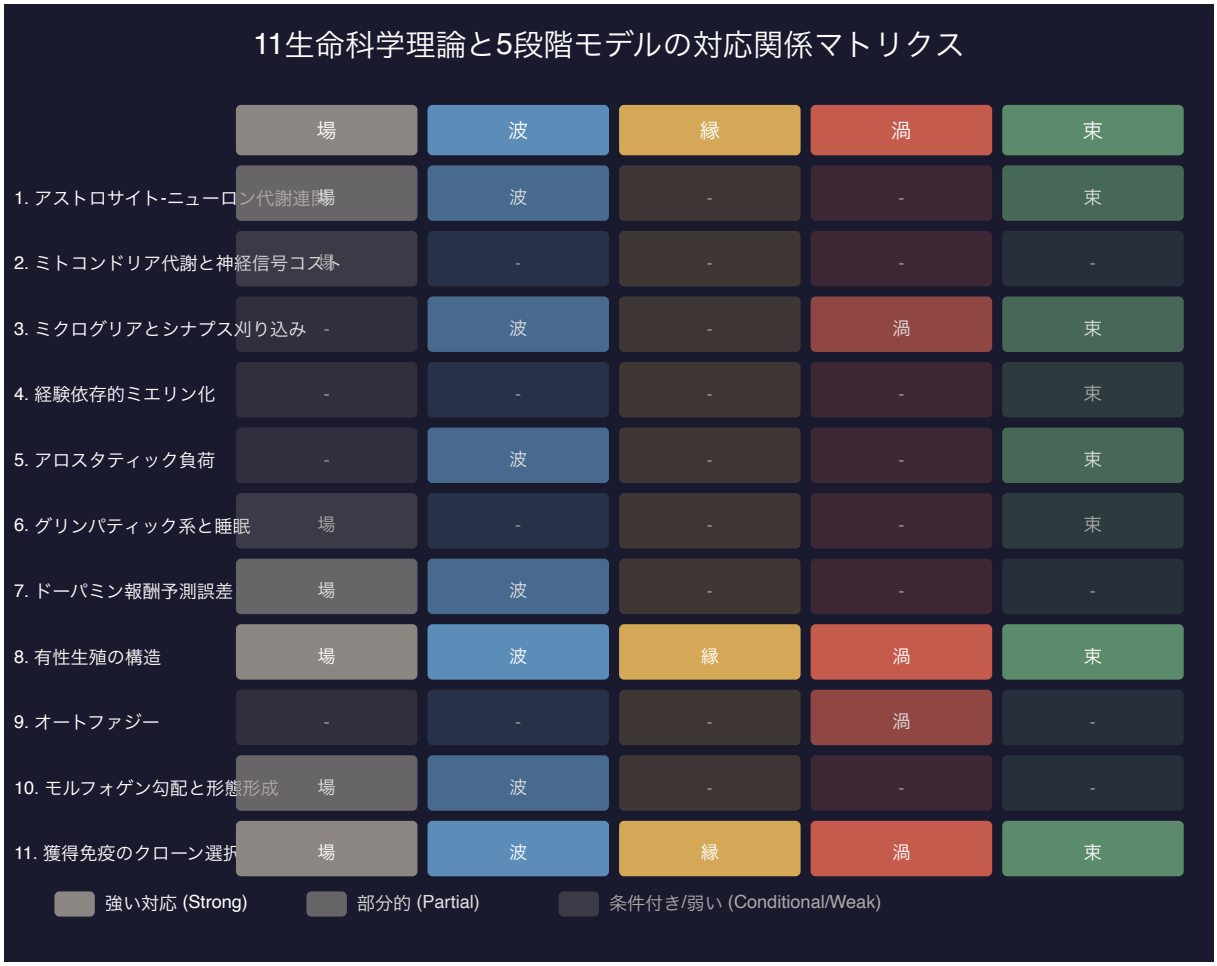

Figure 2: Domain — Theory x 5-Stage Correspondence Matrix

diversity is generated through crossing-over during meiosis and random segregation of gametes; fertilization then fuses two haploid gametes to reconstitute a new diploid individual. Van Valen's (1973) Red Queen hypothesis showed that the co-evolution of hosts and parasites continuously demands genetic diversity. As Maynard Smith (1978) and Otto (2009) noted, sexual reproduction is maintained evolutionarily despite carrying a "twofold cost" relative to asexual reproduction (the cost of producing males, half of the offspring, who do not themselves bear offspring).

- **As fact**: The process of sexual reproduction proceeds from a pool of genetic diversity in the population, through the generation of haploid cells by meiosis, the encounter and sexual selection of gametes, the formation of a new diploid individual through fertilization, and finally stabilization as offspring. This cycle has continued for more than 4 billion years.
- **As reading**: Here, we read an irreversible yet repeating process structure: generation of diversity → selection → fusion → stabilization → cycling back to the next round. The level of similarity is structural — the focus is not on superficial label proximity but on the relational arrangement itself: "from a pool of diversity, through relationship (the encounter of gametes), a new individual emerges." In particular, the encounter and sexual selection of gametes exemplifies the structure of "different things (male and female) meeting at a boundary, with the relationship generating a new existence." The "twofold cost" is a structural feature in which high cost is accepted in order to maintain diversity.
- **As interpretation**: The population (pool of genetic diversity) corresponds to the **Field**; meiosis and sexual differentiation (symmetry breaking) correspond to the **Wave**; the encounter and sexual selection of gametes (different things meeting at a boundary, selected by relationship) correspond to the **Edge**; fertilization (from the fusion of two haploid cells a new diploid individual emerges) corresponds to the **Vortex**; offspring (individuals that stabilize as a new genetic combination) correspond to the **Bundle**. Furthermore, offspring becoming the next generation's population (Field) produces a natural Bundle → Field cycle.
- The most important point that sexual reproduction illustrates is that this 5-stage correspondence occurs without consciousness. For 4 billion years the Field → Wave → Edge → Vortex → Bundle cycle has continued at a level devoid of consciousness. This fact strongly suggests that the 5-stage model is not confined to conscious creation.

Key Entry 2: Clonal Selection of Adaptive Immunity

- Adaptive immunity is a sophisticated defense system that vertebrates have acquired through evolution. According to Burnet's (1957, 1959) clonal selection theory, the body contains in advance a diverse population of lymphocytes bearing varied antigen receptors, and antigens that enter from outside select the matching clone, driving its proliferation and differentiation. Tonegawa (1983) elucidated the molecular mechanism of V(D)J recombination, showing that the RAG-1/RAG-2 enzymes generate receptor diversity through recombination of DNA segments (Schatz et al. 1989). In the thymus, T cells undergo positive and negative selection based on interactions with MHC molecules (Kisielow et al. 1988); in the periphery, somatic hypermutation and affinity maturation proceed in germinal centers.
- **As fact**: The process of adaptive immunity begins with a diversity pool of naïve lymphocytes (the receptor repertoire generated by V(D)J recombination). Antigen stimulation activates lymphocytes; a triple molecular interaction — MHC presentation, TCR recognition, and co-stimulatory signals — is established between antigen-presenting cells (APCs) and T cells. This interaction, following positive/negative selection and competition among clones, leads to the acquisition of effector function and long-term retention as immunological memory.
- **As reading**: Here, we read an irreversible process structure: generation of diversity → relational selection → memory formation. The level of similarity is structural, with particular attention to the structure in which "the triple relationship between APC and T cell (MHC presentation + TCR recognition + co-stimulation) directly controls whether a response succeeds or fails." This triple relationship exhibits a structure in which different types of cells meet at a boundary, and direction is determined by the relational quantity of molecular recognition specificity. Furthermore, affinity maturation in germinal centers is an iterative refinement process — somatic hypermutation → selection of high-affinity clones — in which precision improves through spiral repetition rather than a single pass.
- **As interpretation**: The diversity pool of naïve lymphocytes (the repertoire before selection)

corresponds to the **Field**; activation and proliferation triggered by antigen stimulation corresponds to the **Wave**; the triple molecular interaction between APCs and T/B cells (MHC presentation + TCR recognition + co-stimulation) corresponds to the **Edge**; convergence through positive/negative selection and inter-clonal competition corresponds to the **Vortex**; acquisition of effector function and re-bundling as immunological memory corresponds to the **Bundle**.

- The important point that adaptive immunity illustrates is that the Edge is implemented by the physical quantity of "molecular recognition specificity." This is a distinct implementation lineage from the Edge implementations found in other domains — the time difference at synapses, phase synchrony in neural oscillations, social engagement in polyvagal theory — and suggests that the Edge is not a specific mechanism but a more general structural feature.

Key Entry 3: Morphogen Gradients and Morphogenesis

- Morphogen gradients are a fundamental mechanism for spatially determining cell fate during embryonic development. Wolpert's (1969) positional information hypothesis provided a framework in which cells receive and interpret the concentration of signaling molecules as a "positional value." Driever & Nüsslein-Volhard (1988) showed that the bicoid protein in Drosophila forms a concentration gradient along the anterior-posterior axis, with position specified in a concentration-dependent manner. Turing (1952), meanwhile, showed mathematically that coupled reaction and diffusion of an activator and an inhibitor can generate periodic patterns through self-organization from a uniform state. Kondo & Asai (1995) demonstrated empirically that the striped patterns on fish skin are consistent with the wave-like behavior of this reaction-diffusion model.
- **As fact**: Morphogenesis is described through two lineages. In the first (Wolpert-type), a concentration gradient of a signaling molecule provides positional information to cells, and discrete cell fates are determined by thresholds — a "top-down" mode. In the second (Turing-type), periodic patterns spontaneously emerge from uniform initial conditions through coupled local reaction and diffusion of an activator and an inhibitor — a "bottom-up" mode.
- **As reading**: Here, we read a process structure along the spatial axis: uniform initial condition → diffusion of information → fate bifurcation at threshold → self-organization → fixation of pattern. The level of similarity is process, with particular attention to the transformation structure in which "a continuous gradient (concentration) is converted into discrete fates (cell differentiation) through a threshold." We also note that two independent lineages — the top-down Wolpert type and the bottom-up Turing type — exhibit the same structural transition.
- **As interpretation**: The near-uniform initial condition (the embryo before pattern formation) corresponds to the **Field**; the temporal development of gradient formation or reaction-diffusion corresponds to the **Wave**; threshold interpretation and mutual inhibition (converting a continuous concentration into discrete fates — a relational conversion) corresponds to the **Edge**; sharpening of boundaries and self-organization of domains corresponds to the **Vortex**; the fixed spatial pattern (segmentation, stripe patterning) corresponds to the **Bundle**.
- The most important point that morphogen gradients illustrate is that the 5-stage structure can unfold not only along a time axis but also along a spatial axis. Typically the 5 stages are described as a temporal process, but in morphogenesis "Field → Wave → Edge → Vortex → Bundle" appears as the spatial structuring of the embryo. This finding raises questions about the scope of the 5-stage model's applicability.

Cross-Cutting Patterns

- Across life sciences, the following four patterns were confirmed
- The most prominent pattern found in this survey is that the 5-stage structure appears at levels devoid of consciousness
- This finding suggests the possibility that the 5-stage model describes not only "conscious creation" but also a more general "pattern of structural change"
- Microglial synaptic pruning (section 5.4) and autophagy (section 5.5) are processes of "removing and decomposing unnecessary elements for reuse"

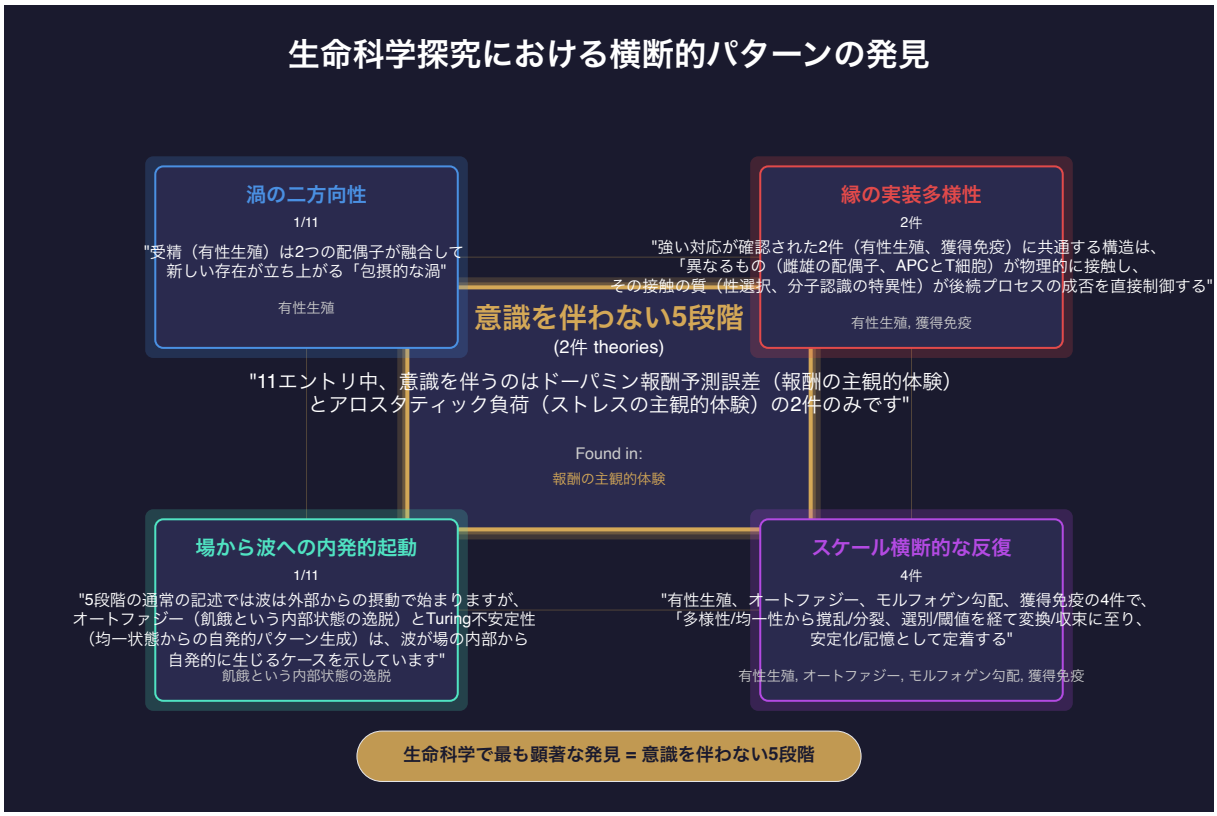

Figure 3: Domain — Cross-Cutting Patterns Diagram

Open Questions

- **Positioning of destructive vortices**: "Decomposition" in microglial pruning and autophagy runs in the reverse direction of the Vortex (which "emerges as a coherent whole"). Whether to accommodate this as a variant of the same structure or to classify it as qualitatively different remains unresolved.
- **Failure modes of the 5 stages**: Allostatic load describes a "failure" of the 5 stages — a pathological state in which Wave is not processed at the Edge and accumulates. Whether this failure mode is a variant contained within the 5 stages or a separate process external to them requires theoretical clarification.
- **Spatial 5 stages**: Morphogen gradients show the 5 stages unfolding along a spatial axis. If the 5 stages are applicable not only to temporal processes but also to spatial patterns, what dimensions can support them? Whether dimensions other than time and space are applicable remains uninvestigated.
- **Significance of the 5 stages without consciousness**: The majority of phenomena surveyed are processes devoid of consciousness. The appearance of the 5-stage structure in life processes without consciousness raises the question of whether the 5 stages describe "the process of creation" or a more general "pattern of structural change." This distinction determines the scope of the theory.

Conclusion

- This survey confirmed that life sciences, as a whole, form a domain with well-supported structural correspondence to the 5-stage model
- The three greatest contributions of life sciences to this survey are as follows
- These findings carry implications in the direction of expanding the scope of applicability of the 5-stage model
- The findings of this survey are distributed across the well-supported to hypothesis temperature range